Agreement-Stratified Evaluation of Deep Learning for Anatomical Landmark Recognition in Upper Gastrointestinal Endoscopy

Presented by

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Outline

Introduction

- Upper GI endoscopy is the primary screening route for gastric cancer; its yield depends on inspecting the whole mucosal surface.
- The Systematic Screening protocol fixes a photographic route over 22 landmarks — a grid of 4 gastric walls by 6 depth stations.
- A model that recognises the landmark in each frame can act as a real-time coverage monitor and flag an unvisited region.
- Published systems report macro F1 near 85–88, and the task is generally treated as solved [6].
  - But every one of those figures is measured only on frames all four experts labelled identically — 60.2% of this corpus.

The reported number describes the easy fraction of the task.

Problem Identification

60.2% of the corpus is unanimous — and that is all the literature scores on.

- 39.8% is discarded before any result is reported.
- The discarded images are not noise: 50.96% of conflicts put two experts on different walls of the same station.
- A deployed system meets the contested fraction continuously, and gets no signal that its validation excluded those frames.
  - Operating accuracy is unknown on exactly the images where a second reader would help most.

Research Questions

RQ	Pre-registered primary endpoint
RQ1	ceiling-normalised macro F1 gap, S-unanimous minus S-majority (Phase 3B)
RQ2	annotator-marginalized macro F1, C2 minus C3, on the pooled contested stratum
RQ3	within-stratum Spearman rho between model uncertainty and annotator vote entro
RQ4	expected anatomical error distance, C4 minus C2, at lambda = 1, on the pooled
RQ5	number of modality-independent gates separating the sound from the unsound cor

One primary endpoint per question, each frozen before the analysis that tests it ran.

Background — the published baselines

Model / label construction	Params	Macro F1
ConvNeXt-Large	200 M	88.25 +/- 0.22
ConvNeXt-Tiny	28 M	approx. 85
ResNet-152	60 M	85.28 +/- 0.27
ConvNeXt-Tiny, fellow-agreement labels	28 M	87.05 +/- 0.21
Best single annotator as target	-	84.82 +/- 0.23
Human expert band	-	77.47 - 84.82

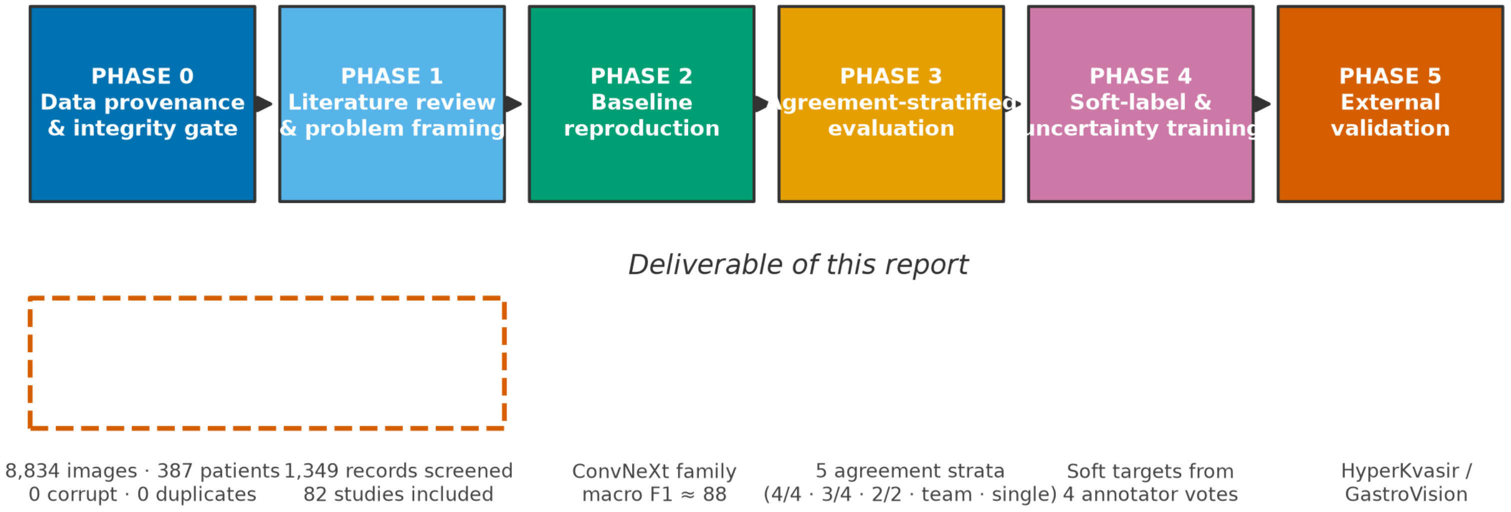
Changing only how the four labels are combined moves F1 by 2.2 points — more than the gap between architecture families [6].

The Gap — counted, not asserted

Reporting dimension	Mentioning	%
Population description (age / sex)	0/68	0.0%
Calibration of predicted probability	7/68	10.3%
Predictive uncertainty	18/68	26.5%
External validation or shift	29/68	42.6%
How the reference standard was built	29/68	42.6%

Of the 29 studies that engage with how their reference standard was built, only **5** also mention whether their probabilities can be believed.

Methodology — eight gated phases



Each phase freezes its hypotheses and verdict rules before any model runs; the generating script refuses to overwrite them.

Data and corpus audit

Measurement	Value
Images / patients / classes	8,834 / 387 / 23
Annotators	FG1, FG2, G1, G2
Decode integrity	8,834 decoded, 0 missing, 0 corrupt
Near-duplicate pairs examined	39,015,361
Cross-split duplicates (calibrated rule)	0
Fleiss' κ / Krippendorff's α / Gwet's AC1	0.7475 / 0.7475 / 0.7475

Chosen because it releases the individual annotators' labels. Without them, no endpoint in this design exists.

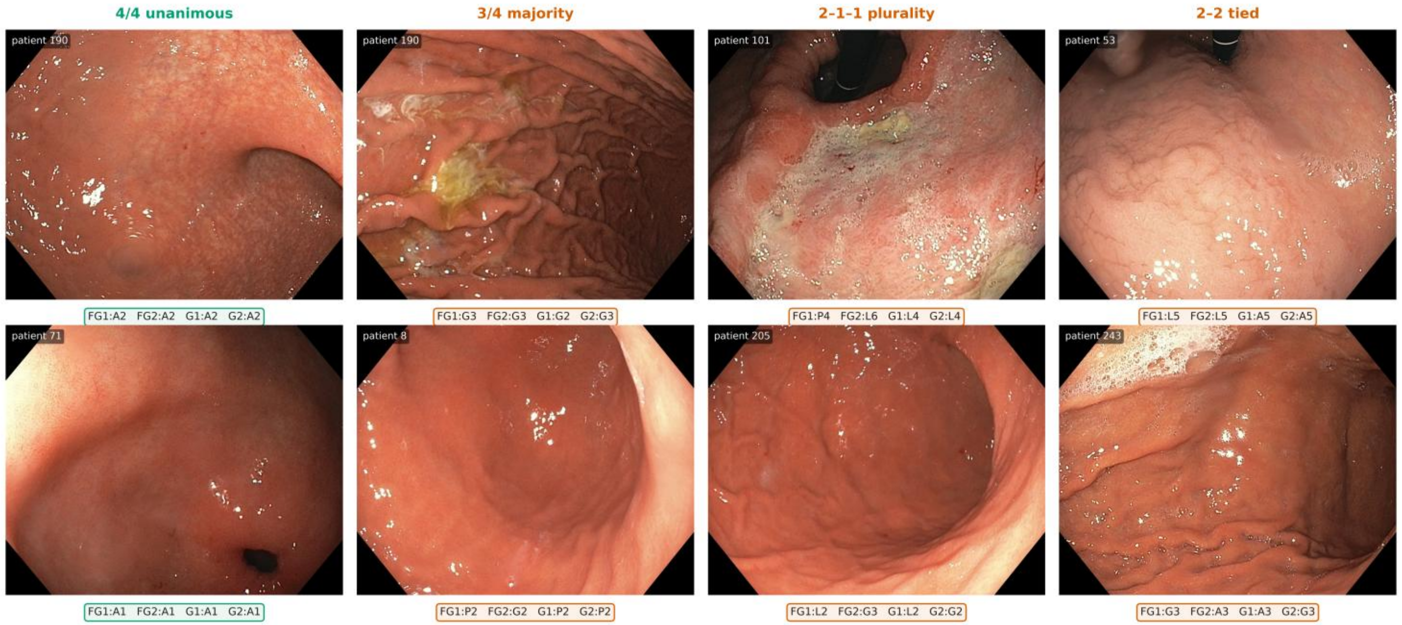
Pre-processing — and the augmentation we refuse

- Resampled once to 224x224 with Lanczos and cached, so every arm in every later phase reads pixel-identical inputs.
- Normalised on this cohort's own channel statistics, not the ImageNet constants — endoscopic illumination is red-dominant and they differ by up to 0.127 in the mean.
- NO horizontal or vertical flips, and no large rotations.
  - The label encodes a gastric WALL — anterior, posterior, lesser curvature, greater curvature. A horizontal flip maps one wall onto the appearance of another and silently relabels the image.
- What remains is a mild scale and aspect jitter (crop scale 0.85-1.0) and a photometric jitter (brightness, contrast, saturation +/-0.2, hue +/-0.02) — neither moves the anatomy.

An augmentation that destroys the target is not regularisation.

Sample dataset — agreement falling, left to right

Sample images from the corpus, with the label each of the four annotators gave



Left to right, agreement falls. Every image is a legitimate frame from the screening protocol; the disagreement is about which landmark it shows, not about image quality.
Published evaluations on this corpus score only the leftmost column. Source: data/phase3_test_manifest.csv.

Published evaluations score only the leftmost column. Input: a 224x224 frame. Output: one of 23 classes with a confidence.

Baseline reproduction — the validity gate

Metric	Value
Macro F1, three-seed mean	83.92 (95% CI 81.47 - 86.20)
Published target	85 +/- 1.5, fixed before training
Difference	-1.08 points
Accuracy / weighted F1	84.39 / 84.35
Expected calibration error	9.27%
Per-seed macro F1	83.67, 83.14, 84.94
Pre-registered verdict	PASS

PASS — the pipeline behaves as the published one did. Everything after this is attributable to the design, not to a bug.

Two attainable ceilings — and why they differ

Stratum	Panel ceiling	Leave-one-out oracle	Difference
Unanimous 4/4	1.0000	1.0000	+0.0000
Majority 3/4	0.7423	0.7423	+0.0000
Plurality 2-1-1	0.4456	0.4492	+0.0037
No majority	0.4023	0.5452	+0.1429
Contested (pooled)	0.6476	0.6697	+0.0221

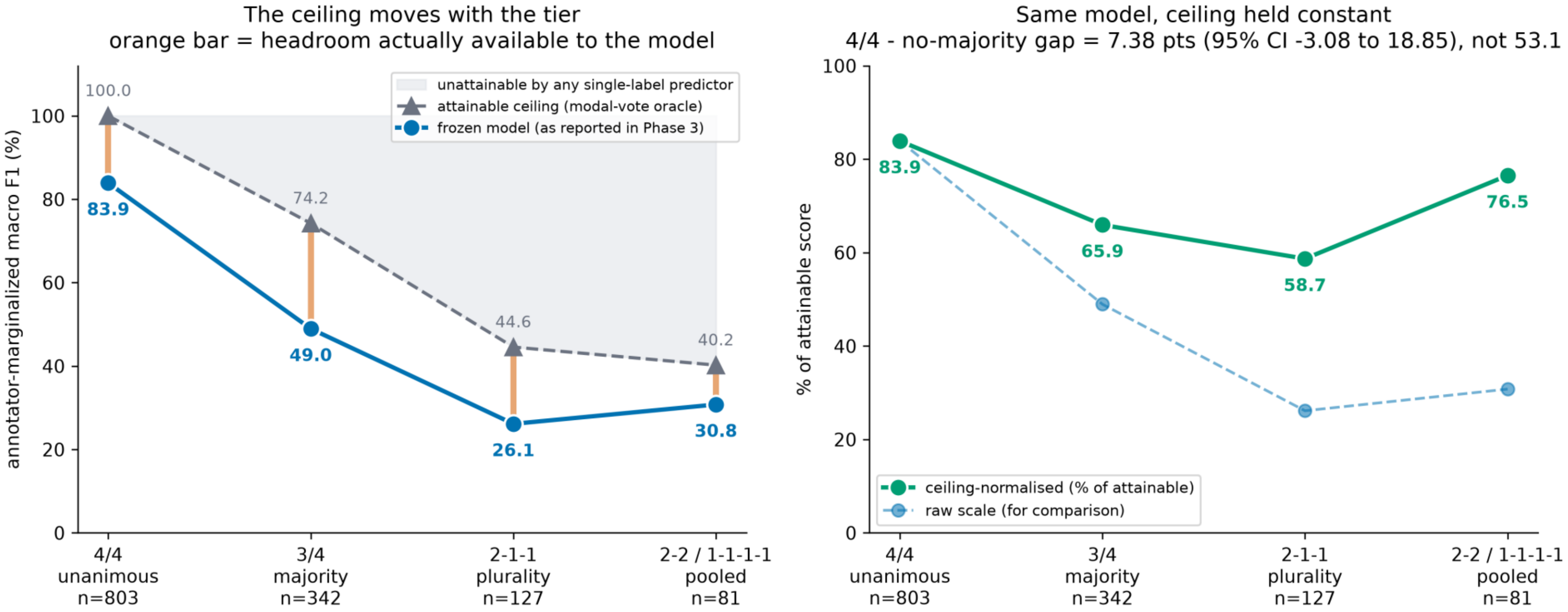
They coincide where the mode survives removal and diverge by 14.29 points where it does not — tie multiplicity, not a bug. RQ1 uses only the strata where they agree.

RQ1 — performance across agreement strata

Stratum	n	Macro F1	Ceiling	ECE %
Unanimous 4/4	803	83.9	1.0000	9.2
Majority 3/4	342	48.9	0.7423	38.2
Plurality 2-1-1	127	26.2	0.4456	56.4
No majority	81	30.8	0.4023	49.1

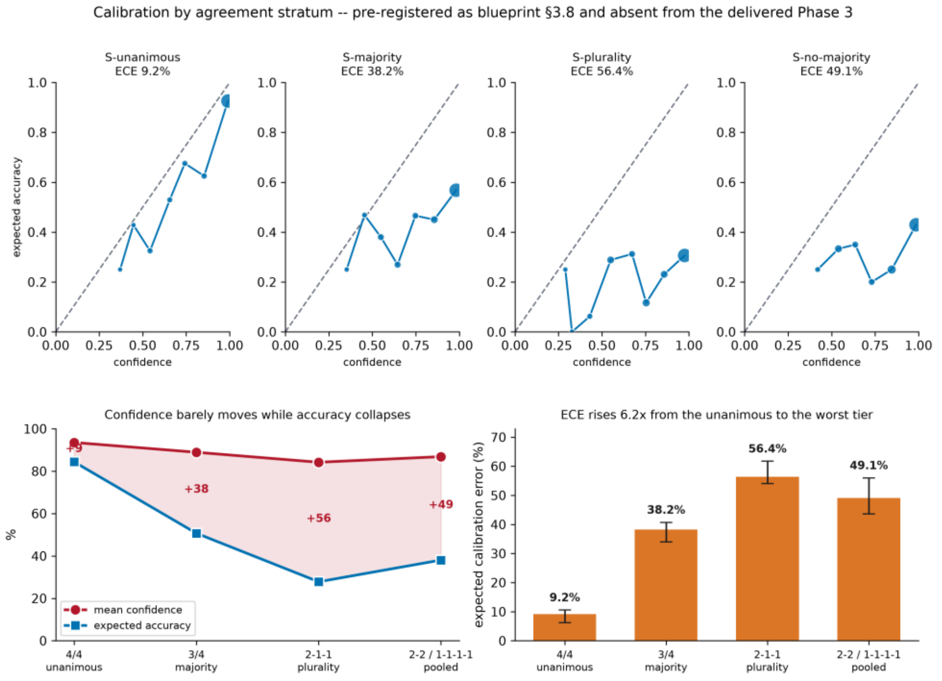
Raw macro F1 falls 83.9 → 26.2. But the ceiling falls with it.

RQ1 — the ceiling moves, so normalise by it



Ceiling-normalised unanimous-minus-majority gap: 17.98 points (95% CI [12.49, 24.37]), not the 34.97 the raw scores suggest.

The calibration collapse



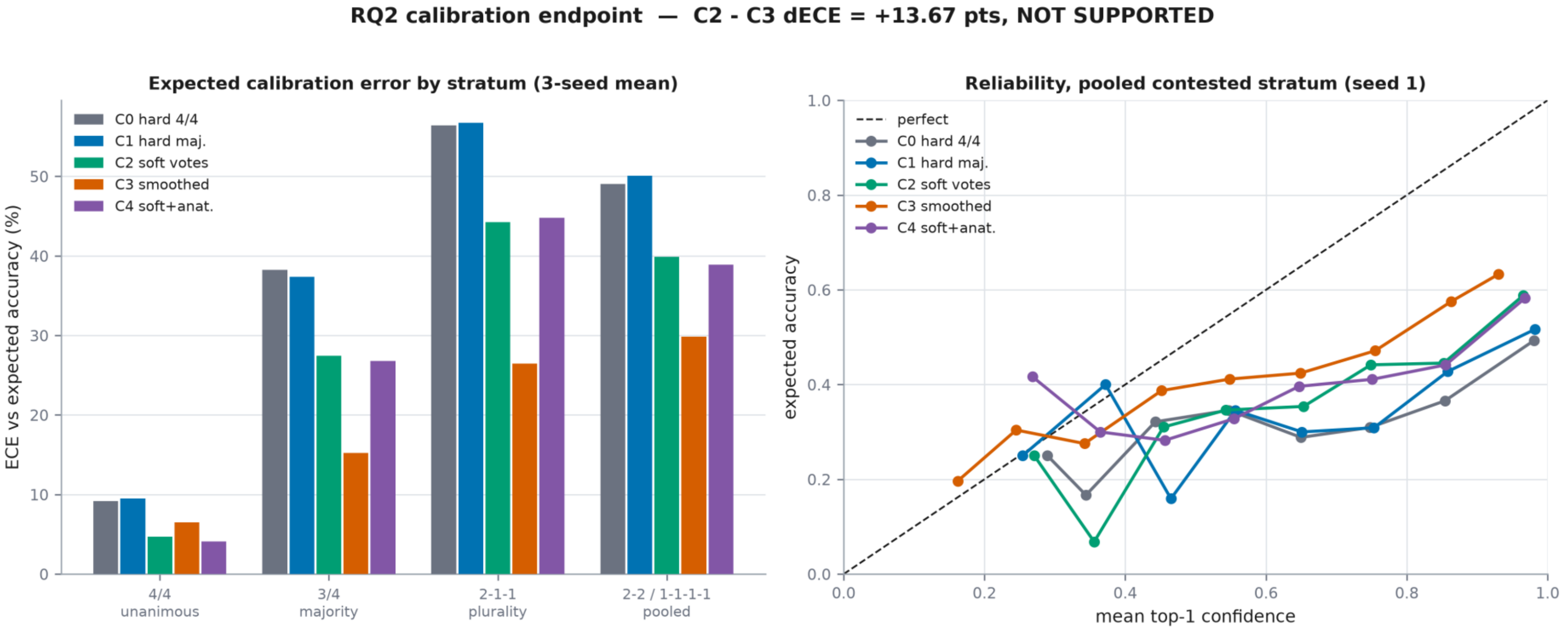
ECE 9.2% → 56.4%. Confidence falls 9.34 points while accuracy falls 56.57. The durable finding of the thesis.

RQ2 — five target constructions, one cohort

Arm	Target construction
C0	hard label, 4/4 cohort
C1	hard majority label
C2	vote proportions (soft target)
C3	hard + matched smoothing (CONTROL)
C4	vote proportions + anatomical penalty

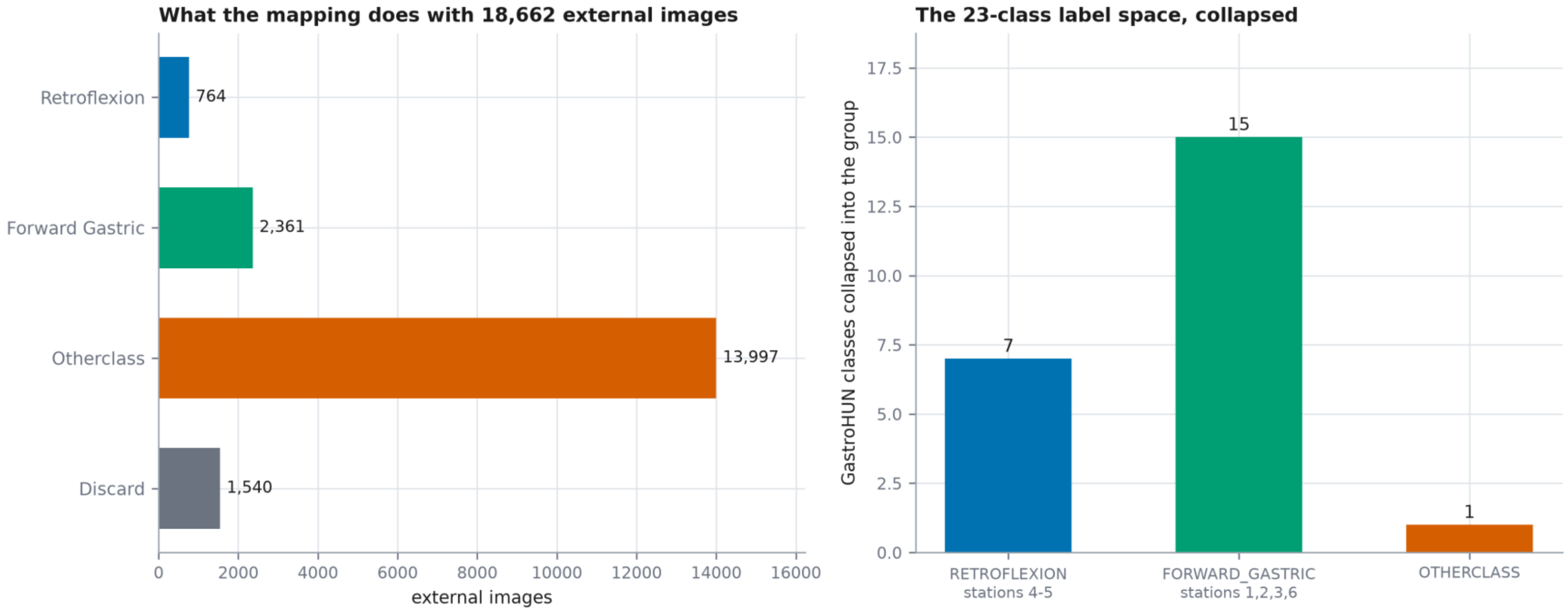
ε = 0.07529 derived to match the probability mass the soft target displaces — not set to a convention. Backbone, schedule and selection identical across arms.

RQ2 — the reversal is the finding



Accuracy contrast C2−C3 = 1.40 ([-0.88, 3.68]): NOT RESOLVED. On calibration the generic control WINS on contested images. No target construction repairs it.

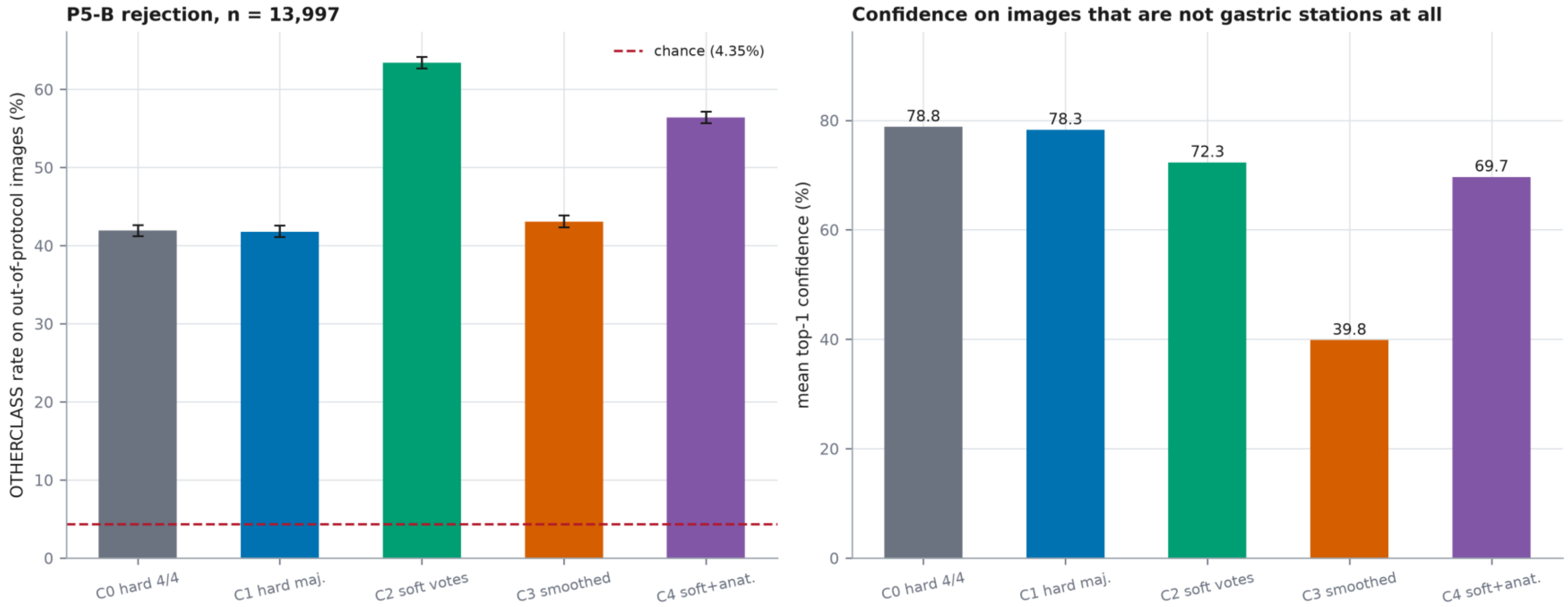
External validation — what the mapping destroys



The wall axis of the GastroHUN label space is unrecoverable from either external corpus, and four of the six stations have no external counterpart.
Phase 5 therefore tests a 2-way anatomical collapse, not 23-way station classification.

Neither external corpus carries the wall axis. A 23-way external validation is not available — the phase was reframed before any image was scored.

Out-of-protocol rejection separates the arms



A model that is wrong AND confident on out-of-protocol input is worse than one that is merely wrong: the right-hand panel is the deployment-relevant quantity.

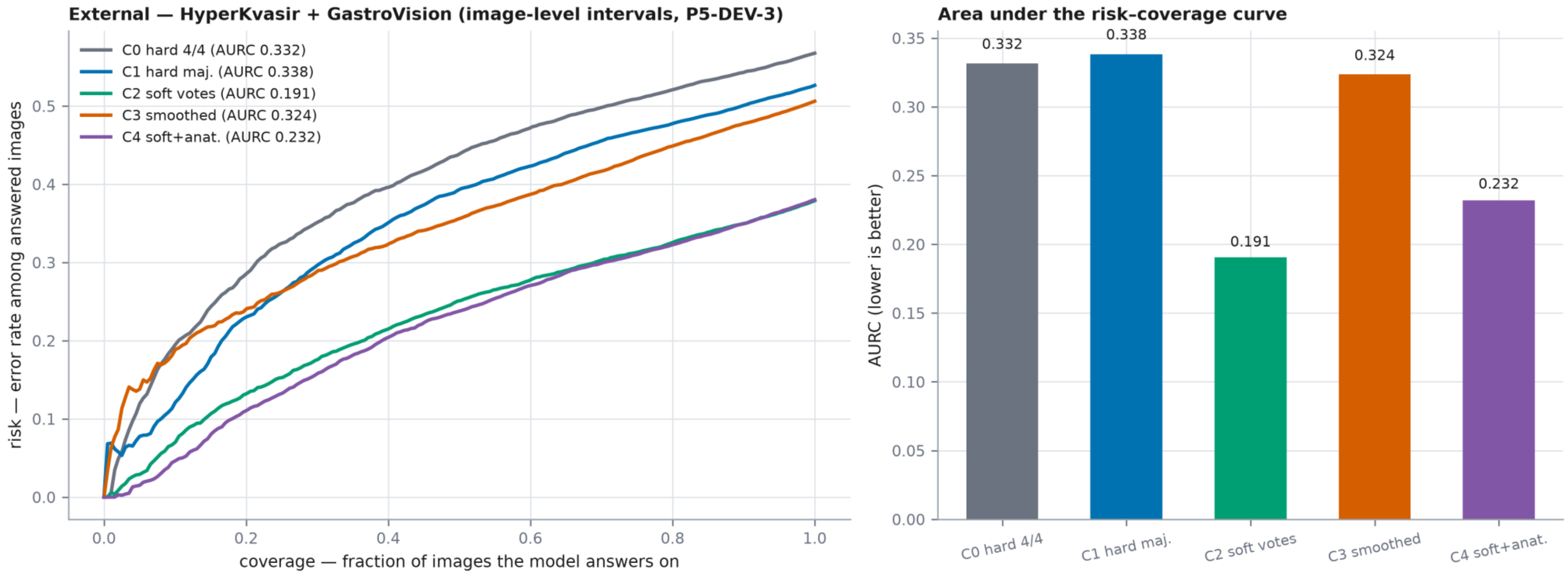
Invisible internally: GastroHUN's test split holds almost no out-of-protocol images. This is the argument for a second centre stated as a measurement.

RQ3 — the human comparator, and the oracle that bounds it

Stratum	Held-out annotator	Model (C2)	Modal-vote oracle	Headroom
Unanimous 4/4	1.0000	0.8514	1.0000	n/a
Majority 3/4	0.4893	0.5301	0.7423	16%
Plurality 2-1-1	0.1436	0.2668	0.4492	40%
No majority	0.2506	0.3135	0.5447	21%
Contested (pooled)	0.3902	0.4575	0.6700	24%

Out-predicts an individual annotator — but NOT the panel's own modal vote, on any stratum. It is not evidence of superior anatomical judgement.

Selective prediction — where the arms finally separate



External AURC 0.1906 for C2 against 0.3235 for C3 — decisive, where every internal endpoint failed to separate them.

RQ5 — a negative control on our own audit protocol

Phase 0 gate applied to both corpora — the audit protocol discriminates

	Retired dataset (Peptic Ulcer_Dataset.xlsx)	GastroHUN
Real patients, verifiable provenance	✗	✓
Named ethics approval + informed consent	✗	✓
Independent expert labels retained	✗	✓
Inter-annotator agreement quantifiable	✗	✓
Labels independent of the features	✗	✓
Official patient-level splits published	✗	✓
Signal present above the majority floor	✗	✓
Peer-reviewed data descriptor	✗	✓
Reusable under an open licence	✗	✓
Complete demographic metadata	✗	✗
Per-class test set adequately powered	✗	✗

Retired dataset: permutation test $p = 0.7622$ — no measurable association between features and target.
GastroHUN: Fleiss $\kappa = 0.748$ across four independent experts on 8,834 images.

PARTIALLY SUPPORTED — gates separate the corpora, but only 1 of 4 fatal defects is caught by any gate, and only incidentally. Well-formedness is not viability.

Is any of this a ConvNeXt artefact?

- The target-construction contrast was re-run on EfficientNet-B0 (4.0 M parameters against ConvNeXt-Tiny's 28 M) [74].
- The training script imports the Phase 4 code and rebinds only the model constructor, so any difference is architectural.
- 3 of 3 claims replicate: the accuracy null, the calibration reversal, and the model's position between annotator and oracle.
  - What this does NOT cover: confusion geometry, attribution stability, and the external endpoints. Those remain single-architecture and are named in the threat table.

The strongest remaining objection, tested — and only partly closed. We say which part.

Threats to validity, ranked

Threat	Severity
Single backbone for geometry, attribution, external	Medium
Compute-bound epoch cap on 8/12 runs	Medium
Contested strata are small (n = 342, 127, 81)	Medium
External intervals are image-level	Medium
Four annotators bound the human comparator	Medium
Strata defined by the same agreement that scores the human	Medium
No age or sex in the corpus	Low

Ranked by how much each could change a conclusion — not by how easy each is to answer.

What could be deployed, and under what conditions

- Nothing here supports deploying an autonomous landmark classifier.
- What it supports is a completeness-checking assistant that ABSTAINS.
- The operating characteristic is the external risk–coverage curve: AURC 0.1906 for C2 against 0.3235 for the internally-best arm.
  - A deployment would have to select on that endpoint, monitor it at the new centre, and route declined images to a human rather than scoring them.

The recommendation is about abstention, not about a better loss — because no target construction repaired calibration.

Conclusions — answers to the five questions

RQ	Verdict under multiplicity
RQ1	EXCLUDES 0; HOLM SURVIVES
RQ2	CONTAINS 0
RQ3	NOT ESTIMABLE
RQ4	CONTAINS 0
RQ5	NO INTERVAL

Three of five returned unresolved or not-estimable. Each was a pre-registered test against a matched control with its precision target met — that is what makes them findings.

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Full list: 86 references, all generated from the review's extraction table.

Thank You

Questions and comments welcome